

WALMART SALES ANALYTICS

Project Overview

The Walmart Sales Analytics Dashboard is an interactive developed using Streamlit and MySQL. It enables users to analyze sales performance through KPIs, charts, and interactive filters, helping identify business trends and support data-driven decisions.

Business Problem

Businesses collect large volumes of sales data, making it difficult to identify trends and performance indicators. This dashboard simplifies analysis by presenting key metrics and interactive visualizations in one place.

Dataset

Walmart Sales

<https://www.kaggle.com/datasets/mikhail1681/walmart-sales>

Features

- Store: Store number
- Date: Sales week start date
- Weekly_Sales: Sales
- Holiday_Flag: Mark on the presence or absence of a holiday
- Temperature: Air temperature in the region
- Fuel_Price: Fuel cost in the region
- CPI: Consumer price index
- Unemployment: Unemployment rate

Tech Stack

Technology	Purpose
Python	Programming
MySQL	Database
SQLAlchemy	Database connection
Pandas	Data analysis
Plotly	Interactive charts
Streamlit	Dashboard

SQL Analysis

- Which store has the highest total sales?
- Which weeks had the highest and lowest sales?
- How do holiday weeks affect sales?
- Which stores perform consistently well?
- What is the average weekly sales per store?
- Does unemployment affect sales?
- Does fuel price affect sales?
- Does temperature affect sales?
- Which month has the highest sales?

Skills Demonstrated

- SQL querying
- Database design
- Data cleaning
- Data visualization
- Dashboard development
- Business analysis
- Python programming

Results and Findings

- Highest-performing Stores
- Best-Weekly Sales per store
- Relationship between (fuel price, temperature, weekly holidays) and weekly sales
- Monthly sales trends